Thermodynamic description of the Al–Cu system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Cu binary system as calculated with pycalphad from the SGTE dataset AlCu-02Mie.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → β + (Cu), lies at 1036.7 °C.

1. Introduction

The Al–Cu system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic rich, peritectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of J. Miettinen, Metall. Mater. Trans. A, 33A, 1639-48(2002). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Al melts at 660.3 °C. The melting point of Cu reproduced by the dataset is 1084.6 °C.

2. Phase diagram and invariant reactions

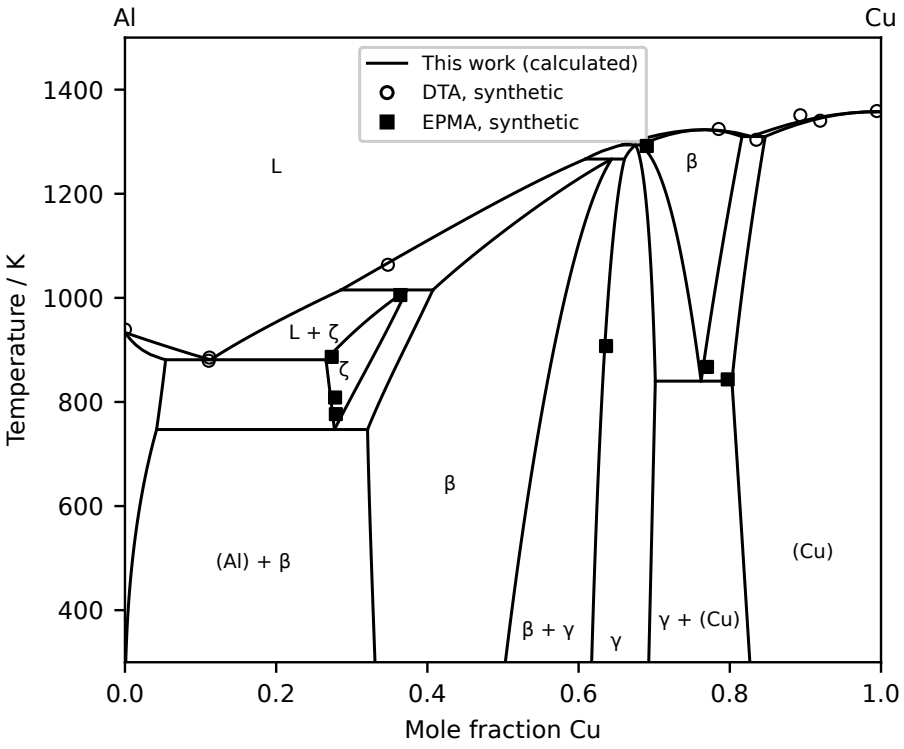


Fig. 1. Calculated Al–Cu phase diagram with synthetic experimental-style points.

The eutectic reaction L → β + (Cu) is calculated at 1036.7 °C, with L at x(Cu) = 0.829. We obtain 1020.6 °C for the peritectic reaction (L + β → γ), where L has x(Cu) = 0.658. The dataset places a peritectic (L + γ → β) at 993.5 °C; the composition of L is a mole fraction of 0.609. We obtain 742.1 °C for the peritectic reaction (L + β → ζ), where L has a mole fraction of 0.286. We obtain 607.9 °C for the eutectic reaction (L → (Al) + ζ), where L has a mole fraction of 0.113. We obtain 566.9 °C for the eutectoid reaction (β → γ + (Cu)), where β has a mole fraction of 0.762. The eutectoid reaction ζ

→ (Al) + β is calculated at 473.9 °C, with ζ at a mole fraction of 0.277. β melts congruently at 1049.9 °C and x(Cu) = 0.765.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Al–Cu).

Reaction	Type	T (K) this work	T exp.[b]	Ref.	at.% Cu, 1	at.% Cu, 2	at.% Cu, 3
L → β	congruent melting[a]	1323.1	–	–	76.5	76.5	–
L → β + (Cu)	eutectic	1309.9	1305.9	[5]	82.9	81.6	84.7
L + β → γ	peritectic	1293.8	–	–	65.8	68.2	67.5
L + γ → β	peritectic	1266.7	1268.7	[5]	60.9	66.1	64.4
L + β → ζ	peritectic	1015.2	1013.2	[4]	28.6	40.7	37.3
L → (Al) + ζ	eutectic	881.1	876.1	[3]	11.3	5.4	26.6
β → γ + (Cu)	eutectoid	840.0	843.0	[5]	76.2	70.1	80.3
ζ → (Al) + β	eutectoid	747.0	743.0	[4]	27.7	4.2	32.1

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.
[a] Two-phase equilibrium at a temperature extremum; both phases have the same composition.
[b] Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Al–Cu).

	Structure			Stability range / °C		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	at.% Cu	at T / °C
(Al)	cF4	A1	Cu	< 27	660.3	0.0–5.4	607
ζ	hP2	A3	Mg	473.9	742.1	26.6–32.6	607

Table 2 (continued).

	Structure			Stability range / °C		Homogeneity range	
Phase	**Pearson**	**Strukturb.**	**Prototype**	**from**	**to**	**at.% Cu**	**at T / °C**
β	cI2	A2	W	< 27	1049.9	35.2–76.3	577
γ	–	–	–	< 27	1020.6	61.7–69.3	27
(Cu)	cF4	A1	Cu	< 27	1084.6	80.3–100.0	567

According to Table 2, β remains stable up to 1049.9 °C (congruent melting). The solubility of Cu in (Al) reaches a mole fraction of 0.054 at 607.9 °C.

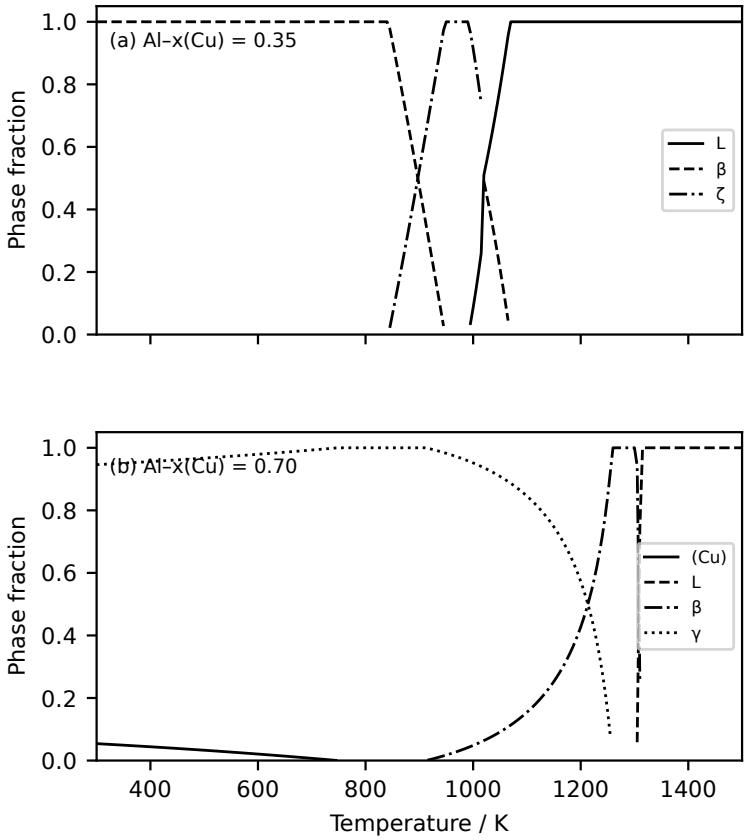


Fig. 2. Calculated equilibrium phase fractions versus temperature for Al–x(Cu) = 0.35 and x(Cu) = 0.70.

4. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its minimum, −17.57 kJ/mol, at x(Cu) = 0.600. The calculated enthalpy of formation of γ at 298.15 K is −20.31 kJ/mol per mole of atoms. Its heat capacity at room temperature is 24.40 J/(mol K). For the equiatomic liquid at 1450 K, the Gibbs energy of mixing is −22.08 kJ/mol.

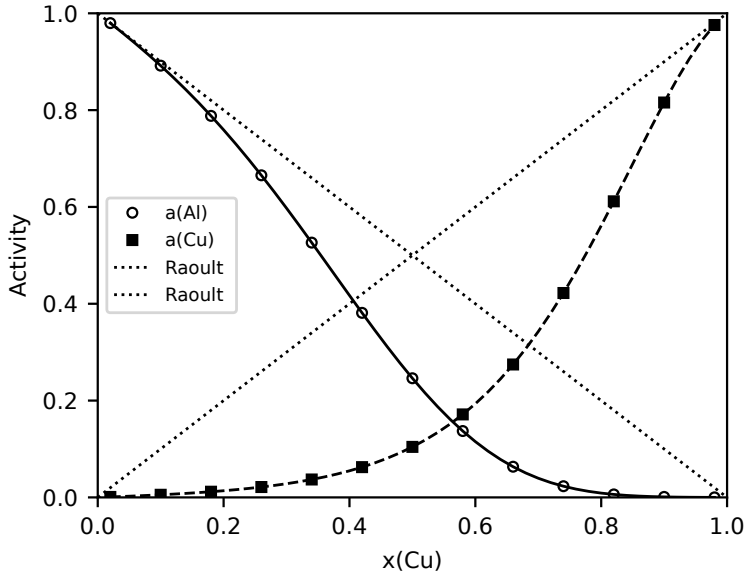


Fig. 3. Calculated activities in the liquid at 1450 K (reference: pure liquids).

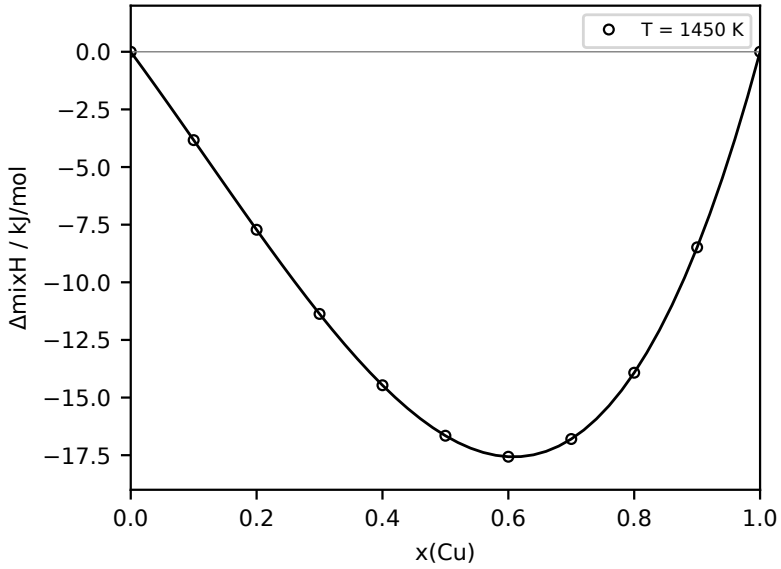


Fig. 4. Calculated mixing enthalpy of the liquid at 1450 K.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Al–Cu).

		298.15 K, per mole of atoms		
Phase	**at.% Cu**	**ΔfH / J mol⁻¹**	**ΔfS / J mol⁻¹ K⁻¹**	**Cp / J mol⁻¹ K⁻¹**
γ	66.6	-20310 ± 500	1.27	24.40
β	46.9	-18662 ± 500	1.73	24.36
ζ[a]	28.9	-11096 ± 1000	3.14	24.34

[a] Not stable at 298.15 K in this description; properties of the metastable phase.

Table 4. Integral and partial mixing properties of the liquid at 1450 K (Al–Cu).

at.% Cu	ΔmixH / J mol⁻¹	ΔmixG / J mol⁻¹	a(Al)	a(Cu)
10	-3832	-7530	0.8918	0.0054
20	-7721	-12989	0.7592	0.0138
30	-11372	-17245	0.5974	0.0283
40	-14466	-20318	0.4172	0.0549
50	-16655	-22076	0.2461	0.1043
60	-17567	-22304	0.1152	0.1934
70	-16798	-20721	0.0396	0.3426
80	-13922	-16961	0.0089	0.5613
90	-8484	-10510	0.0010	0.8157

Reference states: pure liquid Al and Cu at 1450 K; per mole of atoms.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] J. Miettinen, Metall. Mater. Trans. A, 33A, 1639-48(2002).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.